



Multi-Site Company Network Design and Implementation

Using Cisco Packet Tracer for Secure and Scalable Network Infrastructure

Project Overview

This project focuses on the comprehensive design and implementation of a secure and scalable multi-site company network utilizing Cisco Packet Tracer. The primary goal is to establish a robust network infrastructure that supports two distinct branches connected via a central main router, ensuring efficient and secure communication.

Core Objectives

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Secure & Scalable Network

Building a robust network infrastructure supporting two branches via a central router
- 

OSPF Routing Configuration

Configuring dynamic routing protocol for efficient packet forwarding across sites
- 

DHCP & DNS Services

Setting up automated IP assignment and name resolution services
- 

VLAN Implementation

Implementing VLANs for departmental segmentation and broadcast domain control
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Access Control Lists

Establishing ACLs for granular traffic control and network security
- 

NAT & Port Security

Deploying NAT for IP conservation and security enhancements

Network Topology and Architecture

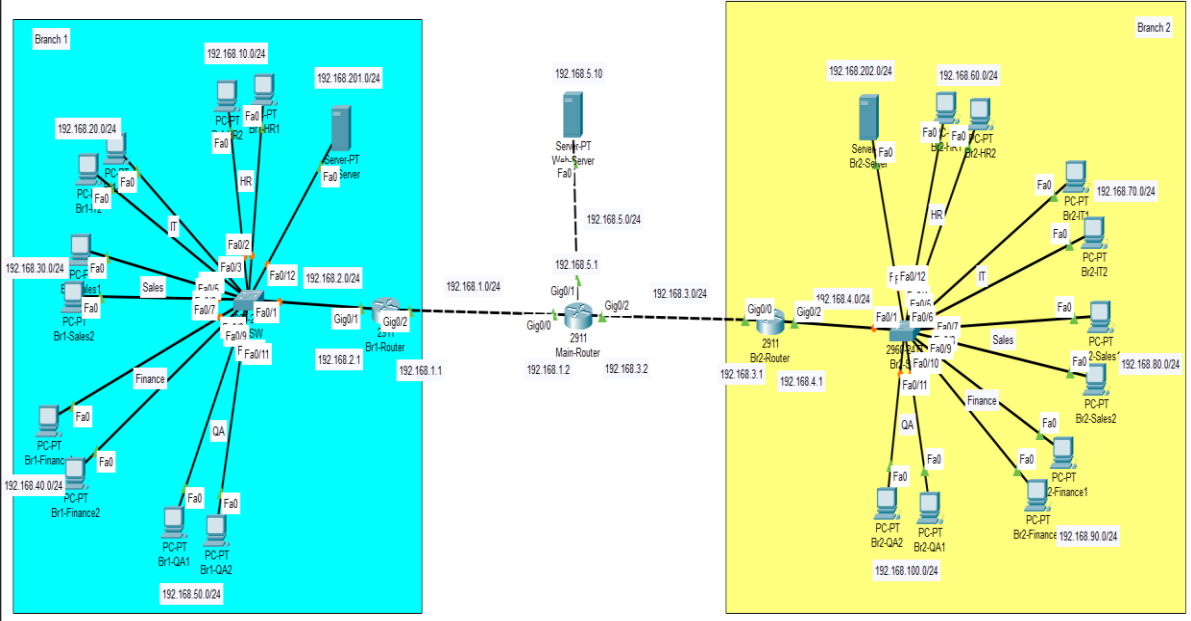
Dual-Branch Topology

The network uses a dual-branch topology with Branch 1 and Branch 2 connected via a Main Router. This facilitates inter-branch communication and centralized management.

Three-Tier Hierarchical Model

- **Access Layer**
Connects end devices to the network. Implemented through VLANs for departmental segmentation.
- **Distribution Layer**
Interconnects different VLANs and enforces security policies. Handles routing and filtering.
- **Core Layer**
Provides high-speed backbone connectivity between branches. Uses OSPF for optimal path selection.

Network Topology Diagram



Structured IP Addressing Plan

A well-structured IP addressing scheme is critical for efficient network operation. The following table outlines the IP addressing plan for both branches and core WAN links.

BRANCH	DEPARTMENT	VLAN ID	SUBNET	GATEWAY
BRANCH 1	HR (LAN)	10	192.168.10.0/24	192.168.10.1
	IT (LAN)	20	192.168.20.0/24	192.168.20.1
	Sales (LAN)	30	192.168.30.0/24	192.168.30.1
	Finance (LAN)	40	192.168.40.0/24	192.168.40.1
BRANCH 2	HR (LAN)	60	192.168.60.0/24	192.168.60.1
	IT (LAN)	70	192.168.70.0/24	192.168.70.1
	Sales (LAN)	80	192.168.80.0/24	192.168.80.1
	Finance (LAN)	90	192.168.90.0/24	192.168.90.1
WAN/CORE	Links	IP Ranges	Gateways	
Br1-WAN1: 192.168.1.0/24, Br2-WAN2: 192.168.3.0/24, Central Web: 192.168.5.0/24				

VLAN Implementation and Inter-VLAN Routing

VLAN Implementation

VLANs segment the network into distinct broadcast domains, enhancing security and managing traffic efficiently. Each department is assigned a unique VLAN ID.

Key Benefits:

- ✓ Enhanced security through network segmentation
- ✓ Reduced broadcast traffic
- ✓ Improved network performance

VLAN Assignments

Branch 1 HR VLAN 10	Branch 1 IT VLAN 20
Branch 2 HR VLAN 60	Branch 2 IT VLAN 70

Router-on-a-Stick Configuration

Inter-VLAN communication is facilitated using a single physical interface with multiple sub-interfaces.



Example Configuration:

```
interface g0/1
no ip address
shutdown
interface g0/1.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
```

Dynamic Routing Protocol

OSPF (Open Shortest Path First) is a link-state routing protocol that ensures optimal path selection and rapid convergence in case of network changes.

Key Benefits:

- ✓ Optimal path selection
- ✓ Rapid convergence
- ✓ Scalability

Configuration Commands:

```
$ router ospf 1
network 192.168.0.0 0.0.255.255 area 0
```



DHCP Services

- ✓ Automates IP address assignment across all departments
- ✓ Centralized servers in each branch (VLAN 201 & 202)

Challenge:

DHCP broadcasts don't cross VLAN boundaries

Solution:

Configured **ip helper-address** on router sub-interfaces

```
ip helper-address 192.168.201.10
```

Points to DHCP server for all departmental VLANs



DNS Services

- ✓ Enables name resolution for network resources
- ✓ Allows access using domain names instead of IPs
- ✓ Simplifies access to central web server

Benefits:

- ★ Enhances user experience
- ★ Simplifies network resource access

NAT and Port Security Implementation



Network Address Translation (NAT)

Purpose & Benefits

- **IP Conservation:** Internal hosts use single public IP for external access
- **Security:** Hides internal private IPs from external networks

NAT Configuration

```
Router(config)# access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.5.0 0.0.0.255
Router(config)# ip nat inside source list 100 interface gigabitEthernet 0/0

Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip nat outside
```

Key Points

- Only traffic to Central Web Server is NATed
- Inter-branch traffic bypasses NAT for security inspection



Port Security

Purpose & Benefits

- **Access Control:** Prevents unauthorized device connections
- **MAC Address Protection:** Mitigates MAC address spoofing

Port Security Configuration

```
Switch(config)# interface range fastEthernet 0/2 - 10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport port-security
Switch(config-if-range)# switchport port-security maximum 1
Switch(config-if-range)# switchport port-security mac-address sticky
```

Violation Mode

Restrict Mode:

- Drops unauthorized traffic
- Logs the violation
- Keeps port active

Access Control Lists Security Policies

Extended ACLs were implemented on the ingress path of each departmental sub-interface, enforcing the principle of Least Privilege to enhance network security.



IT Administrator Access

IT department has full outbound access to all networks



IT Reverse Protection

Blocks non-IT users from initiating connections to IT devices



HR Inbound Protection

HR can initiate traffic, but inbound access is blocked



Finance Total Isolation

No external network allowed to access Finance VLAN



Branch Isolation

HR, Finance, and QA restricted to their own local branch



Cross-Branch Access

IT and Sales can access their respective networks in remote branches

 ACLs implemented on all branch routers

Network Validation and Testing Results

Comprehensive testing procedures were conducted to ensure network functionality and effective enforcement of security policies across all segments.



Intra-Branch Communication

Tests confirmed expected communication patterns within each branch and enforcement of ACLs for departmental isolation.



Inter-Branch Connectivity

Validation of secure communication between Branch 1 and Branch 2 through the central Main Router.



ACL Enforcement

Verification that Access Control Lists properly restrict unauthorized access between departments while allowing legitimate traffic.



Service Accessibility

Testing confirmed DHCP, DNS, and web services are accessible to authorized users across all network segments.



Results Summary

- All network segments successfully established connectivity
- ACL rules properly enforced data locality and security policies
- DHCP services functional across all branches
- NAT and Port Security configurations effective



DHCP Issue – Devices Not Receiving IP Addresses

PCs were unable to receive IP addresses from the DHCP server due to DHCP broadcasts not crossing VLAN boundaries.



Solution: ip helper-address

Configured the `ip helper-address` command on router sub-interfaces to relay DHCP requests from each VLAN to the centralized DHCP server.



Unrestricted Inter-VLAN Routing

By default, inter-VLAN routing allowed all departments to communicate without restrictions, creating a security risk.



Solution: Extended ACLs

Implemented Extended Access Control Lists on departmental sub-interfaces to control and restrict communication between different departments, enforcing the principle of least privilege.

Project Summary and Achievements

This project successfully designed and implemented a secure and scalable multi-site company network utilizing Cisco Packet Tracer. By integrating key networking features, the network achieved its core objectives of secure departmental segmentation, efficient inter-branch communication, and robust external connectivity.

Key Achievements



Comprehensive Network Integration

Successfully integrated VLANs, STP, NAT, ACLs, DHCP, and DNS services to create a cohesive network infrastructure



Enhanced Security Framework

Implemented layered security measures including ACLs, NAT, and Port Security to protect network resources



Seamless Inter-Branch Communication

Established secure connections between Branch 1 and Branch 2 while maintaining departmental isolation



Automated Services Implementation

Deployed DHCP and DNS services for efficient IP address allocation and name resolution



Project Outcome: The practical application of these concepts demonstrated a comprehensive understanding of modern network infrastructure design and implementation, providing a solid foundation for real-world networking challenges.